Backup and Recovery Documentation

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ISIN 325 - Database Security

Table of Contents

Backup Models **3**

Robust Backup 3

Intermediate Backup 5

Simple Backup 6

Cloud Backup 7

Support Protocols **8**

Cloud Protocol 8

System Prioritization 8

Staffing Backup Protocol 8

Recovery Process **9**

On-Premises Recovery 9

Cloud Recovery 9

Metrics **10**

Metrics 10

Impact Analysis 10

Turnover & Change Control **11**

Turnover Procedure 11

Change Control Procedure 11

Change Control Documentation 12

Backup & Restore 13

Identification & Removal of Users - Script 13

Backup and Restore - SSMS Wizard 13

Backup and Restore - Scripts 16

Backup Models

Robust Model

Robust	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
1:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
2:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
3:00 AM	Full	Full	Full	Full	Full	Full	Full
4:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
5:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
6:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
7:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
8:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
9:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
10:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
11:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
12:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
1:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
2:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
3:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
4:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
5:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
6:00 PM	Full	Full	Full	Full	Full	Full	Full
7:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
8:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
9:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
10:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
11:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction

Used For: Transactional systems such as Order Entry, Fulfillment, and Accounting.

Backup Schedule:

- Full backups: 3 AM and 6 PM
- Transaction log backups: Hourly

Estimated Backup Time:

- Full: 2 to 4 hours
- Transactional: ~30 minutes

- Log backups: Under 5 minutes

Retention Policy:
- Daily backups are retained for 14 days
- Weekly backups are retained for 1 month
- Monthly backups are retained for 12 months
- Annual archives are retained for 7 years for compliance

RPO: 1 hour
RTO: 2 hours

Storage Locations:
- Primary backup server on separate storage volume
- Offsite disaster recovery replication
- Encrypted cloud object storage with immutability enabled

Security Controls:
- AES-256 backup encryption
- Role-based access control
- Multi-factor authentication for DBA access

Intermediate Model

Medium	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12:00 AM							
1:00 AM	Full	Full	Full	Full	Full	Full	Full
2:00 AM							
3:00 AM							
4:00 AM							
5:00 AM							
6:00 AM							
7:00 AM							
8:00 AM	Full	Full	Full	Full	Full	Full	Full
9:00 AM							
10:00 AM							
11:00 AM							
12:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
1:00 PM							
2:00 PM							
3:00 PM							
4:00 PM							
5:00 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
6:00 PM							
7:00 PM							
8:00 PM							
9:00 PM							
10:00 PM							
11:00 PM							

Used For: Forecasting, Planning, and other internal legacy systems.

Backup Schedule:

- Full backups: 1 AM and 8 AM
- Transaction log backups: Twice daily

Retention Policy:

- Twice weekly backups for 2 weeks
- Weekly backups for 1 month

RPO: 4 hours
RTO: 4 to 6 hours
Backups are stored locally and replicated to off-site disaster recovery storage.

Simple Model

Simple	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12:00 AM							
1:00 AM							
2:00 AM							
3:00 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
4:00 AM							
5:00 AM							
6:00 AM							
7:00 AM							
8:00 AM							
9:00 AM							
10:00 AM							
11:00 AM							
12:00 PM							
1:00 PM							
2:00 PM							
3:00 PM							
4:00 PM							
5:00 PM	Full						
6:00 PM							
7:00 PM							
8:00 PM							
9:00 PM							
10:00 PM							
11:00 PM							

Used For: Data Warehouse and reporting systems.

Backup Schedule:

- Full backup once per week
- Transaction backups nightly

Retention Policy:

- Weekly backups are retained for 1 month

RPO: 24 hours
RTO: 8 hours

Cloud HR Backup Strategy

The HR system is hosted by a third-party cloud provider.

Cloud Backup Characteristics:

- Daily automated snapshots managed by the provider.
- Geo-redundant storage.
- Vendor-defined SLA for RPO and RTO.
- Weekly encrypted data export retained internally.

Shared Responsibility Model:

- Vendor responsible for infrastructure availability.
- Company responsible for verifying data recoverability and access.
- Quarterly restore verification is required.

Support Protocols

Cloud Protocol

Main Contact - Primary DBA
Backup Contact - Secondary DBA

Primary Contact Method - Phone Number
Secondary Contact Method - Email

System Prioritization

Priority 1:
Order Entry
Accounting
Payroll

Priority 2:
Forecasting
Planning

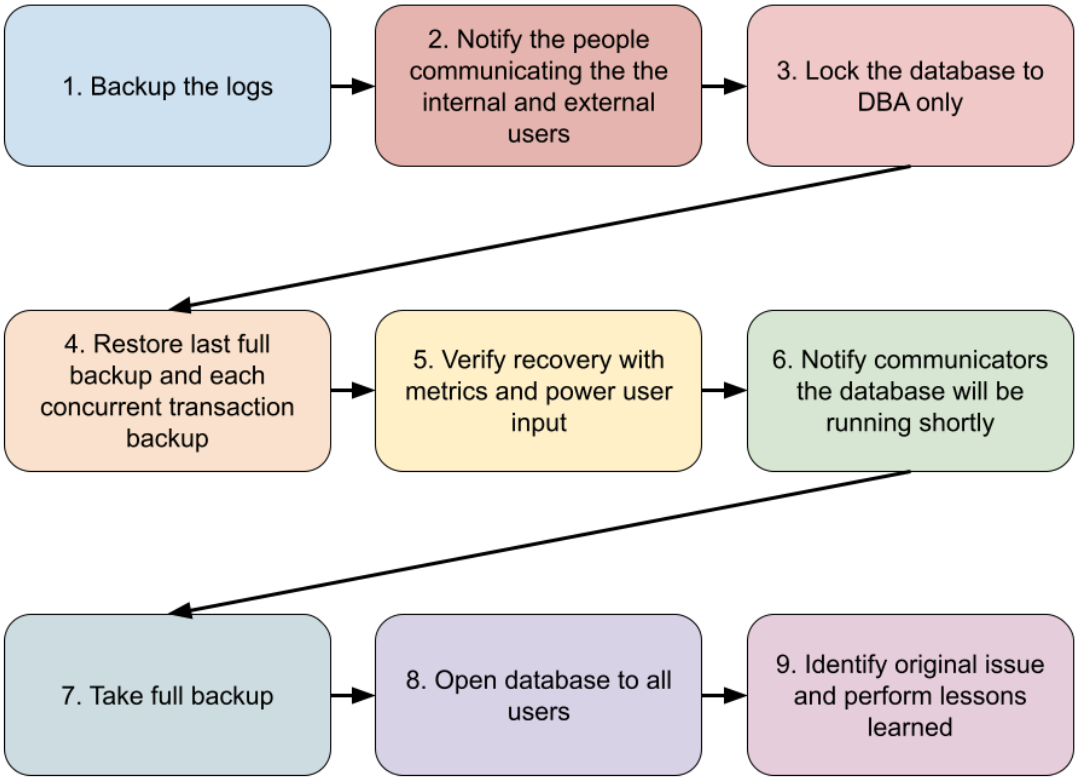
Priority 3:
Data Warehouse

Staffing Backup Protocol

- Shared documentation between all DBAs
- Frequent cross-training
- Weekly documentation updates
- Mandatory credential rotation upon termination
- Annual disaster recovery simulation test
- All service account credentials stored in secure vault with audit logging

Recovery Process

On-premises recovery



Cloud Recovery

1. Contact the cloud vendor support using documented escalation procedures.
2. Confirm recovery point based on vendor SLA.
3. Initiate snapshot restore through the provider.
4. Validate employee and payroll data.

Metrics

- **Full backup duration:** 2 to 4 hours
- **Transaction log backup duration:** Under 5 minutes
- **Database size (Transactional systems):** ~500 GB
- **Database size (Data warehouse):** ~2 TB
- **Average table size:** 1 GB to 50 GB, depending on the system
- **Restore time (Priority 1 systems):** 1.5 to 2 hours
- **Restore time (Data warehouse):** 6 to 8 hours
- **System downtime during incident:**
    - Priority 1: < 2 hours
    - Priority 2: 4 to 6 hours
    - Priority 3: Up to 8 hours
- **Recovery start time:** Within 15 minutes of incident detection
- **Users impacted:**
    - Priority 1: Up to 1500 users
    - Priority 2: ~500 users
    - Priority 3: Limited internal users
- **System outage frequency:** Less than 2 major incidents per year
- **Business impact:**
    - Order Entry downtime: Estimated $10,000 to $25,000 per hour
    - Payroll/HR downtime: High operational and compliance risk
    - Data warehouse downtime: Low immediate financial impact

Impact Analysis

- **Who is affected:** Employees, customers, and finance teams
- **How they are affected:**
    - Inability to process orders
    - Delayed payroll processing
    - Loss of reporting and analytics
- **Business impact includes:**
    - Revenue loss
    - Operational delays
    - Reputational damage

Turnover & Change Control

Turnover Procedure

In the event of DBA turnover or role changes, the following steps will be performed to ensure continuity of operations:

1. Immediately disable all system access for the departing DBA.
2. Rotate all shared credentials, including service accounts and backup systems.
3. Transfer ownership of all backup jobs, scripts, and scheduled tasks.
4. Ensure all documentation is up to date and accessible.
5. Conduct a knowledge transfer session with the remaining DBAs.
6. Review recent backup logs and system status to ensure stability.

All credentials are stored in a secure password vault with audit logging enabled.

Change Control Procedure

All changes to backup and recovery systems must follow a formal change control process.

Change Process Steps

1. Submit a Change Request Form
2. Document the purpose, scope, and risk of the change
3. Obtain approval from the Change Control Board (CCB)
4. Schedule the change during a maintenance window
5. Perform the change and monitor results
6. Document completion and any issues

Change Control Documentation

Change Request Form

Request ID: _____

Date Submitted: _____

Requested By: _____

Change Information

System/Database Affected: _____

Type of Change: _____

Description of Change:

_____

_____

Reason for Change:

_____

Risk & Impact

Risk Level (Low / Medium / High): _____

Potential Impact:

_____

Scheduling

Proposed Implementation Date/Time: _____

Approval

DBA Approval: _____ Date: _____

IT Manager Approval: _____ Date: _____

Change Decision Form

Request ID: _____

Review Date: _____

Change Summary

_____

_____

Decision

Approved / Denied / Deferred

Reason for Decision

_____

Conditions (if approved)

_____

Implementation Window

_____

Approvals

DBA: _____ Date: _____

IT Manager: _____ Date: _____

Backup & Restore

Identification & Removal of Users - Script

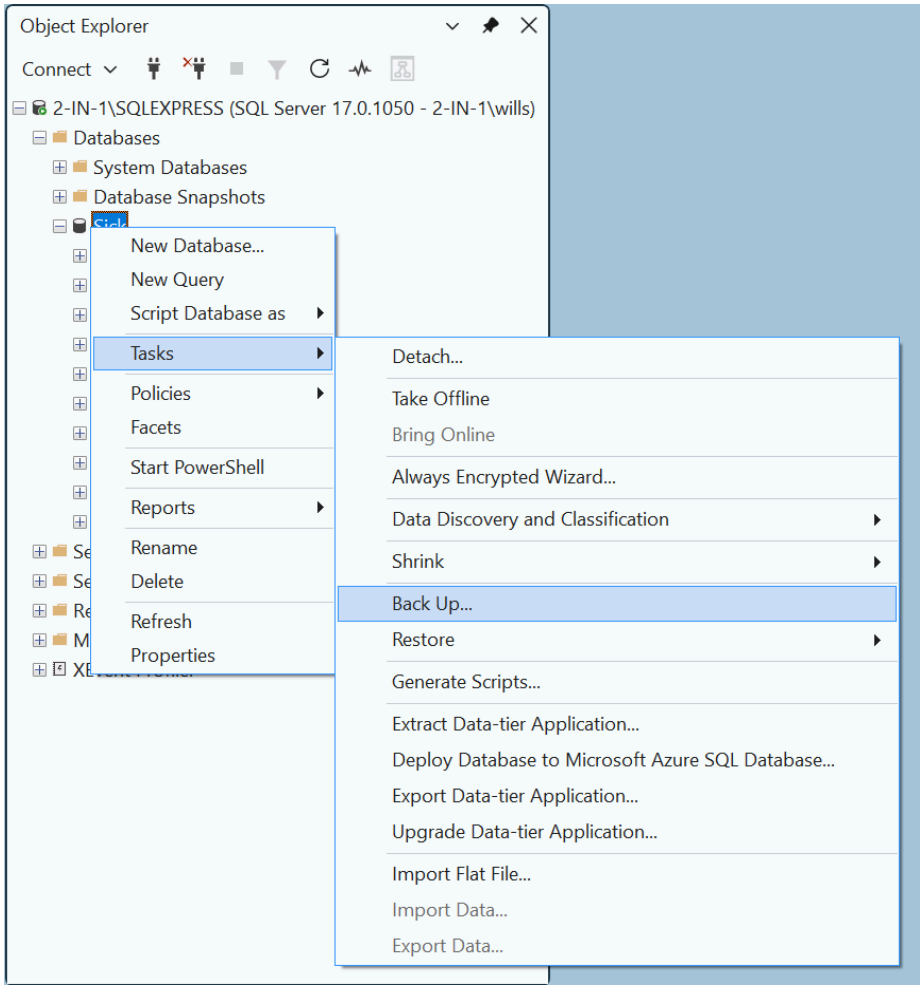
Identification
Sp_who

Removal
Kill __(SPID)

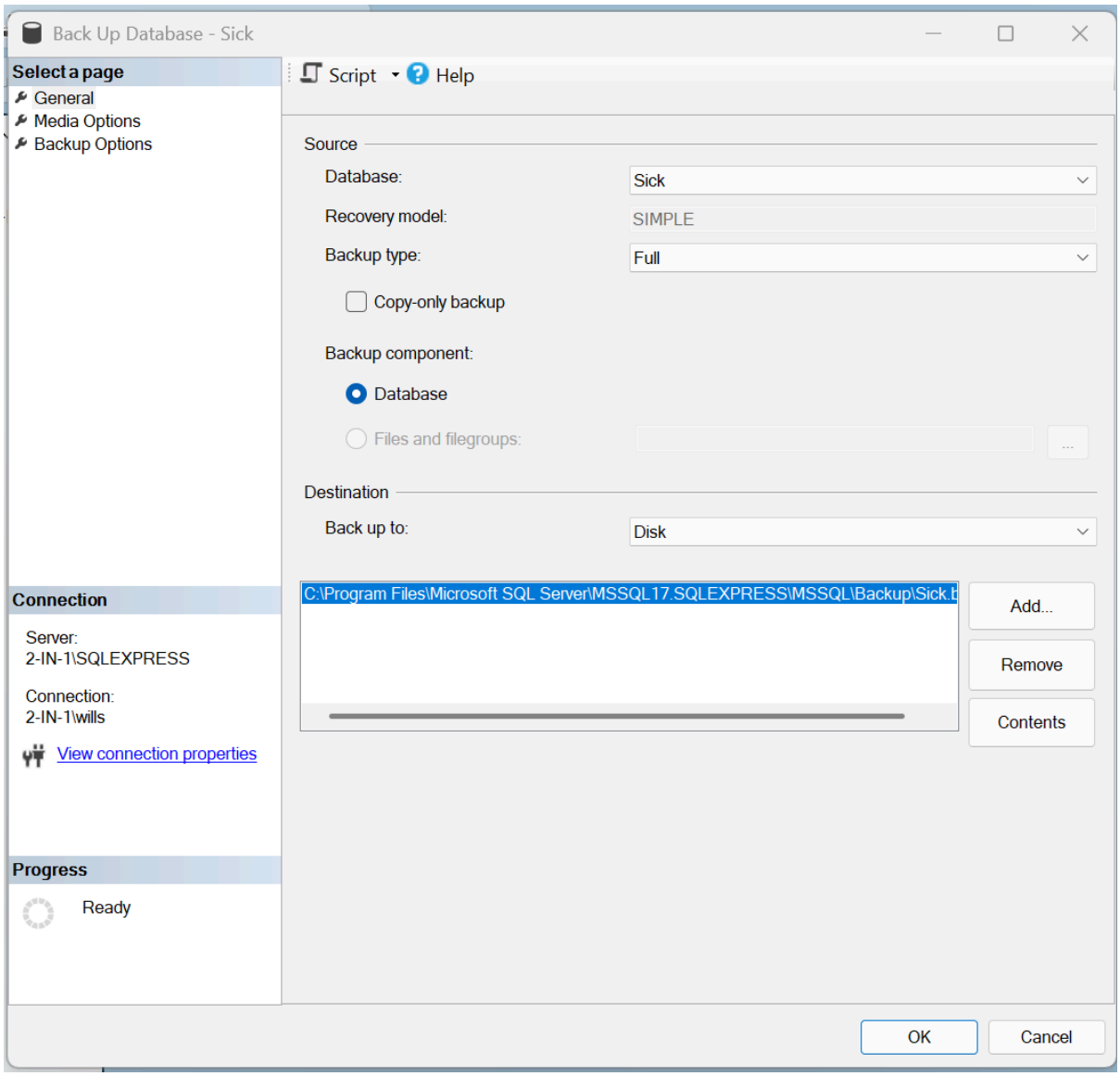
Backup and Restore - SSMS Wizard

Backup

1. Right click on the database, hover over Tasks, then navigate to Back Up…

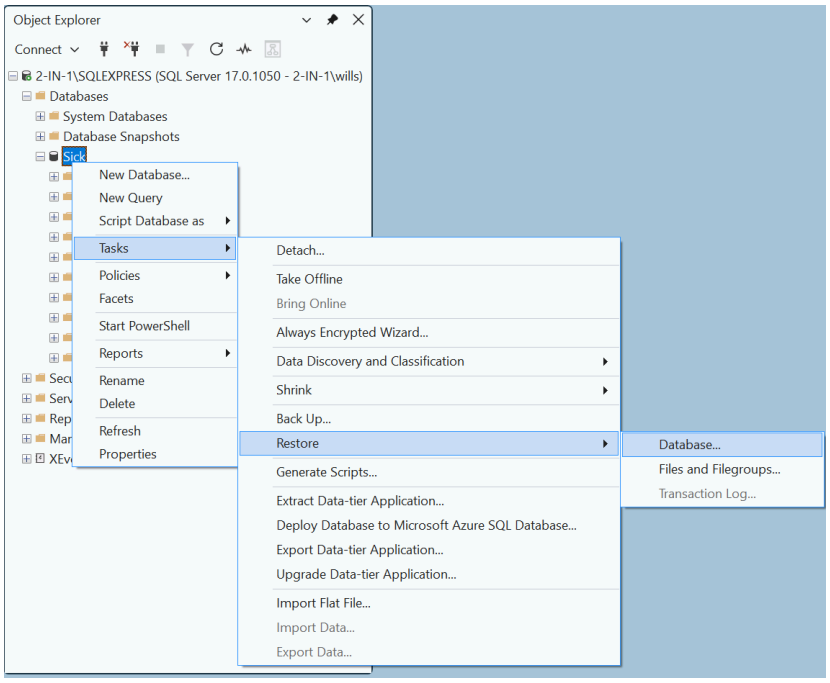


2. Keep the default options and click OK.

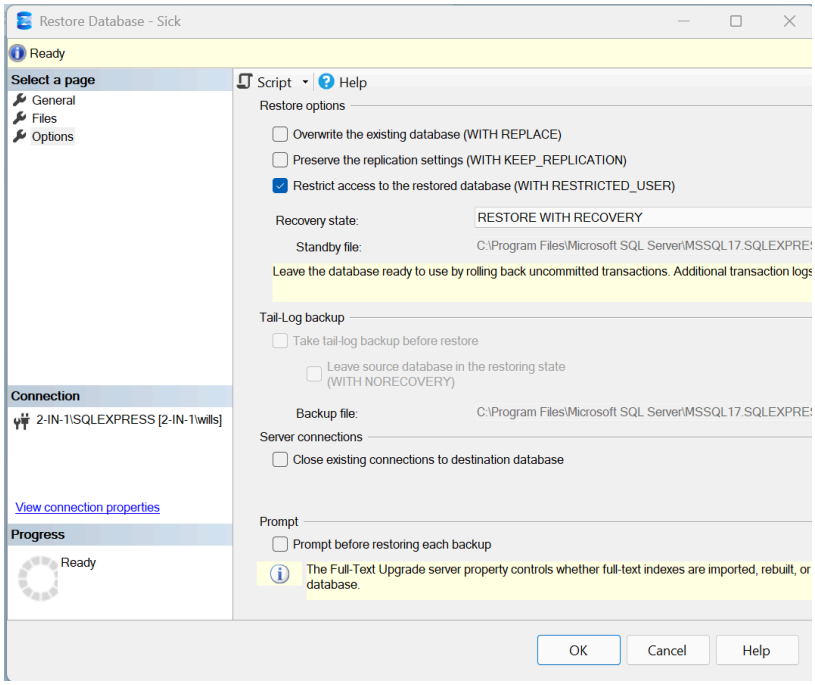


Restore

1. Right click on the database, hover over Tasks, then navigate to Restore and click Database…



2. Now click the options page and select Restrict access to the restored database before pressing OK at the bottom of the page.



Backup and Restore - Scripts

Backup:

```
BACKUP DATABASE Sick
TO DISK = ‘C:\Backups\Sick.bak’
WITH STATS = 10;
PRINT ‘Backup Completed Successfully’
```

Restore:

```
USE master;
GO
ALTER DATABASE sick
SET SINGLE_USER
WITH ROLLBACK IMMEDIATE;
GO
RESTORE DATABASE SICK
FROM DISK = ‘C:Backups\Sick.bak’
WITH REPLACE, RECOVERY, STATS = 10;
GO
ALTER DATABASE Sick
SET MULTI_USER;
GO
```